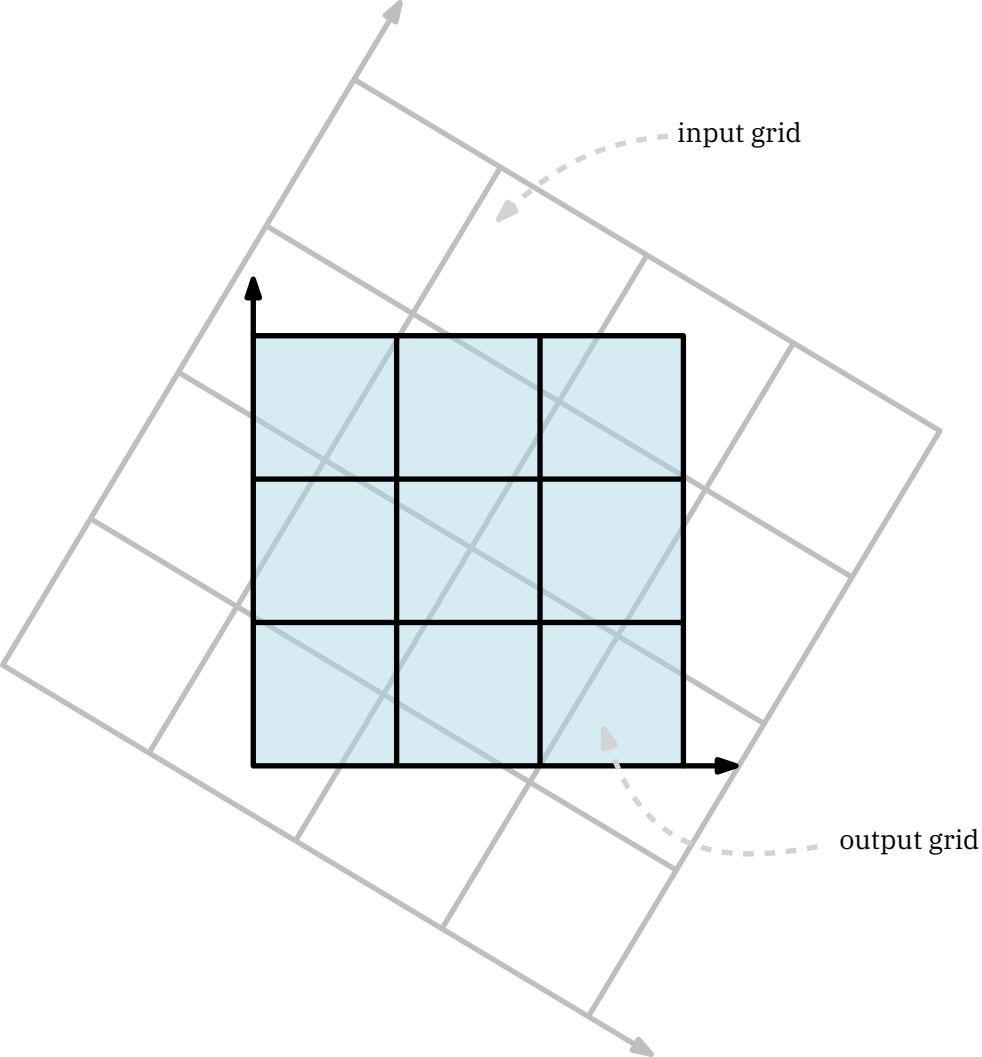
input grid
output grid